

Multiple Choice (10 marks)

- Which of the following lists the hydrides of group-14 elements in order of thermal stability, from lowest to highest?
 - $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$
 - $\text{PbH}_4 < \text{SnH}_4 < \text{CH}_4 < \text{GeH}_4 < \text{SiH}_4$
 - $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$
 - $\text{CH}_4 < \text{PbH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{SiH}_4$
- Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - 23.56 GHz
 - 42.58 MHz
 - 74.34 kHz
 - 13.93 MHz
- Of the following ionic substances, which has the greatest lattice enthalpy?
 - MgO
 - MgS
 - NaF
 - NaCl
- A buffer is made from equal concentrations of a weak acid and its conjugate base. Doubling the volume of the buffer solution by adding water has what effect on its pH?
 - It has little effect.
 - It significantly increases the pH.
 - It significantly decreases the pH.
 - It changes the pH asymptotically to the pK_a of the acid.
- The strongest base in liquid ammonia is
 - NH_3
 - NH_2^-
 - NH_4^+
 - N_2H_4

True / False (10 marks)

Answer True or False

6. True or False: The ^{13}C spectrum of hexane exhibits lines with three distinct chemical shifts due to its symmetrical structure.
7. True or False: The anhydride of $\text{Ba}(\text{OH})_2$ is barium oxide (BaO).
8. True or False: The equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K is 1.71×10^{-5} .
8. True or False: All of the lanthanide elements react with aqueous acid to liberate hydrogen in a vigorous manner.
9. True or False: In the balanced equation, the ratio of MnO_4^- to Mn^{2+} is 3:1.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that influence the chemical shift of protons in NMR spectroscopy, and explain why CH_3Cl exhibits a ^1H resonance approximately 1.55 kHz away from TMS in a 12.0 T magnetic field.
12. Discuss the factors that determine the ratio of line intensities in the EPR spectrum of the t-Bu radical $(\text{CH}_3)_3\text{C}^\bullet$ and explain the significance of these ratios in understanding radical behavior.

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